Introduction to Data Analysis.

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

HEALTH
UNIVERSITY OF UTAH

The “flipped” classroom | (ノ°□°)ノ︵┻━┻

- Every week, we will have lecture materials that are posted online, that way you can watch and re-watch lectures in your own time.
  - Focused on theory and proving concepts.

- In the lab, we will complete assignments that require you to use the concepts discussed in the lecture.
  - Take a quiz based on the lecture/s.
  - Put theory into practice with assignments.
    - Simulations, data manipulation, analysis, graphing, etc.

- The links to the videos, scripts for lab assignments, and all other course materials can be found on the course webpage and on my personal site:

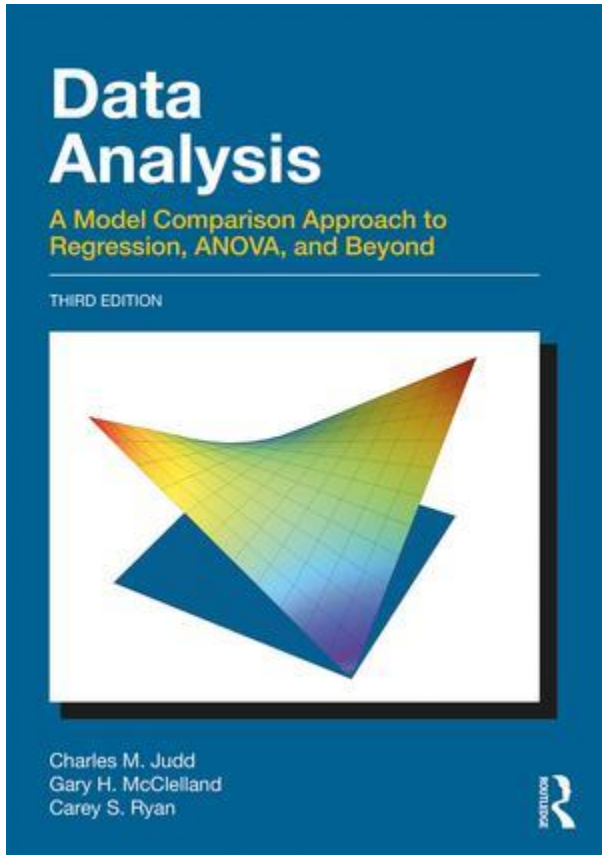
https://sites.google.com/site/lohsekr/data-analysis-i

A Model Comparison Approach

- http://www.dataanalysisbook.com/

- DATA = MODEL + ERROR

  - DATA are our fundamental scores or observations.
  - MODELs are a compact description of representation of the DATA.
  - These MODELs are not perfect, however, and will leave behind residual ERROR.



Different Types of Models

- **Unconditional Models**
  - Estimate one value that best explains the data.
  - On a fair coin, $pr(H) = 0.5$, so having a model $\frac{H}{H+T} = 0.5$ will leave some amount of $ERROR$ depending on how the actual $DATA$ shook out.
    - In small samples the results will be more variable, but in large samples 0.5 might be very good approximation.

$$Y_i = \beta_0 + \varepsilon_i$$

- **Conditional Models**
  - Estimate values contingent on some other factor.
  - Should our estimate of a person's height be modified by their weight? (*Continuous*)
  - Should our estimate of a person's height be modified by their sex? (*Categorical*)

$$Y_i = \beta_0 + \beta_1(x_i) + \varepsilon_i$$

How do we compare models?

- Let's say that one researcher proposes that a person's height is totally unrelated to their weight (**<u>unconditional</u>**):

$$y_i = \beta_0 + 0\ (x_i) + \varepsilon_i$$

- Let's say that another researcher proposes that a person's height is related to their weight (**<u>conditional</u>**):

$$y_i = \beta_0 + \beta_1(x_i) + \varepsilon_i$$

  - We aren't saying exactly what the magnitude of the relationship between height and weight is (β_1) but we are proposing that it is not zero.

How do we compare models?

- If we had access to all of the DATA (i.e., measured everyone to whom our model applies), then we could simply compare the ERRORs in our two different models:

  - *Augmented* Model: $\widehat{y_i} = \beta_0 + \beta_1(x_i)$ and $\sum \varepsilon_i \, , where \; \varepsilon_i = y_i - \widehat{y_i}$

  - *Compact* Model: $\widehat{y_i} = \beta_0 + 0\,(x_i)$ and $\sum \varepsilon_i \, , where \; \varepsilon_i = y_i - \widehat{y_i}$

    $\widehat{y_i} = \beta_0$

How do we compare models?

- If we could measure all of the data, we would simply need to see if ERROR(A) < ERROR(C) to decide if it was the better model.

- However, we face some important issues:
  - **How big of a reduction in ERROR is important?** (Is a 1% reduction in error worth the added complexity.)
  - **We aren't going to get to measure everyone.** (If I'm sampling from a population, how much faith do I have in the reduction in error in my sample?)
  - **Does the absolute or relative error matter?** (If I take a sample of 100 people, the sum of errors will likely be much larger than a sample of 10 people).

How do we compare models?

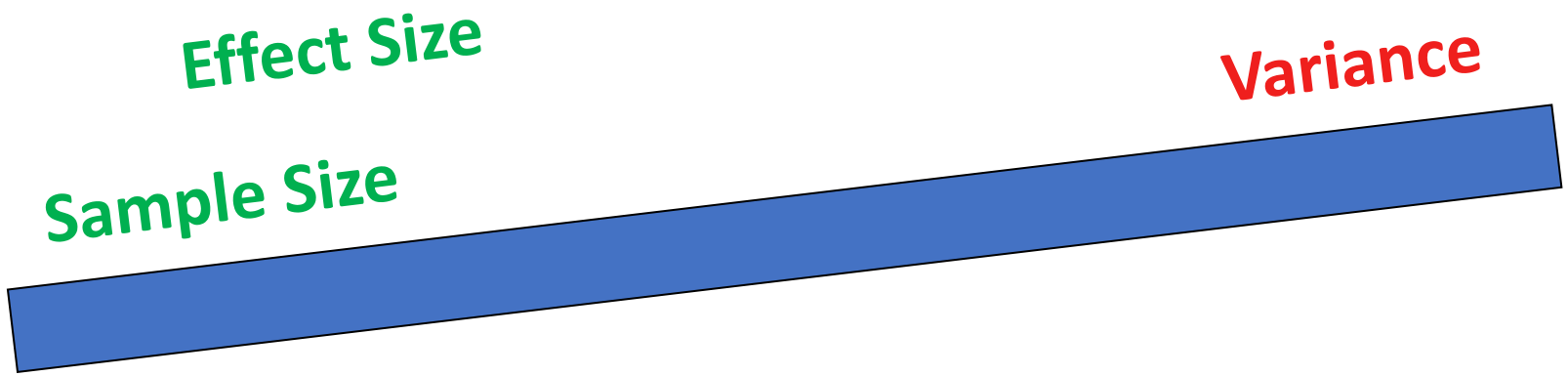
- *<u>Proportional Reduction in Error:</u>*

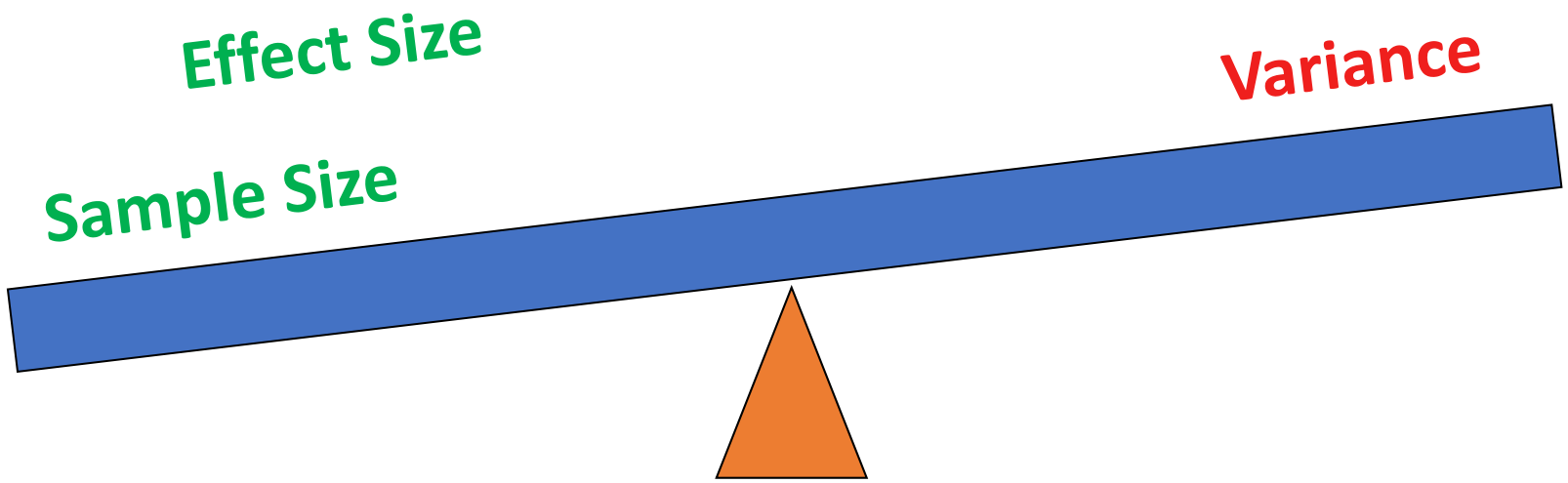
$$PRE = \frac{ERROR(C) - ERROR(A)}{ERROR(C)} = 1 - \frac{ERROR(A)}{ERROR(C)}$$

- So if ERROR(C) = 100 and ERROR(A) = 90,

$$PRE = 1 - \frac{90}{100} = 0.10$$

  - But again, is 10% reduction in error good? If we had all of the data it certainly sounds good… but what about in a sample?


Effect Size
Variance
Sample Size



Everything we do is statistics is an elaboration and specification of this idea!